Table 43. EMI characteristics for f_{HSE} = 8 MHz and f_{HCLK} = 144 MHz

Symbol	Parameter	Conditions	Monitored frequency band	Value	Unit
S_{EMI}	Peak[1]	V_{DD} = 3.6 V, T_A = 25 ° C, LQFP64 package compliant with IEC 61967-2	0.1 MHz to 30 MHz	10	dBµV
			30 MHz to 130 MHz	-1	
			130 MHz to 1 GHz	13	
			1 GHz to 2 GHz	11	
	Level[2]		0.1 MHz to 2 GHz	3.0	-

1. *Refer to the EMI radiated test section of the application note EMC design guide for STM8, STM32 and Legacy MCUs (AN1709).*
2. *Refer to the EMI level classification section of the application note EMC design guide for STM8, STM32 and Legacy MCUs (AN1709).*

5.3.12 Electrical sensitivity characteristics

Based on three different tests (ESD, latch-up) using specific measurement methods, the device is stressed in order to determine its performance in terms of electrical sensitivity.

Electrostatic discharge (ESD)

Electrostatic discharges (a positive then a negative pulse separated by 1 second) are applied to the pins of each sample according to each pin combination. The sample size depends on the number of supply pins in the device (3 parts × (n+1) supply pins). This test conforms to the ANSI/JEDEC standard.

Table 44. ESD absolute maximum ratings

Specified by design and not tested in production.

Symbol	Ratings	Conditions	Packages	Class	Maximum value[1]	Unit
$V_{ESD(HBM)}$	Electrostatic discharge voltage (human body model)	T_A = 25°C conforming to ANSI/ESDA/JEDEC JS-001	All	2	2000	V
$V_{ESD(CDM)}$	Electrostatic discharge voltage (charge device model)	TA = 25°C conforming to ANSI/ESDA/JEDEC JS-002	LQFP 64	C3	1000	
			LQFP 48			
			LQFP 32			
			UFQFPN 48			
			UFQFPN 32			
			UFQFPN 24			

1. *Evaluated by characterization - Not tested in production.*

Static latch-up

The following complementary static tests are required on three parts to assess the latch-up performance:

- A supply overvoltage is applied to each power supply pin.
- A current injection is applied to each input, output, and configurable I/O pin.

These tests are compliant with EIA/JESD 78E IC latch-up standard.

Table 45. Electrical sensitivities

Symbol	Parameter	Conditions	Class
LU	Static latch-up class	T_A = 130°C conforming to JESD78	Level II A